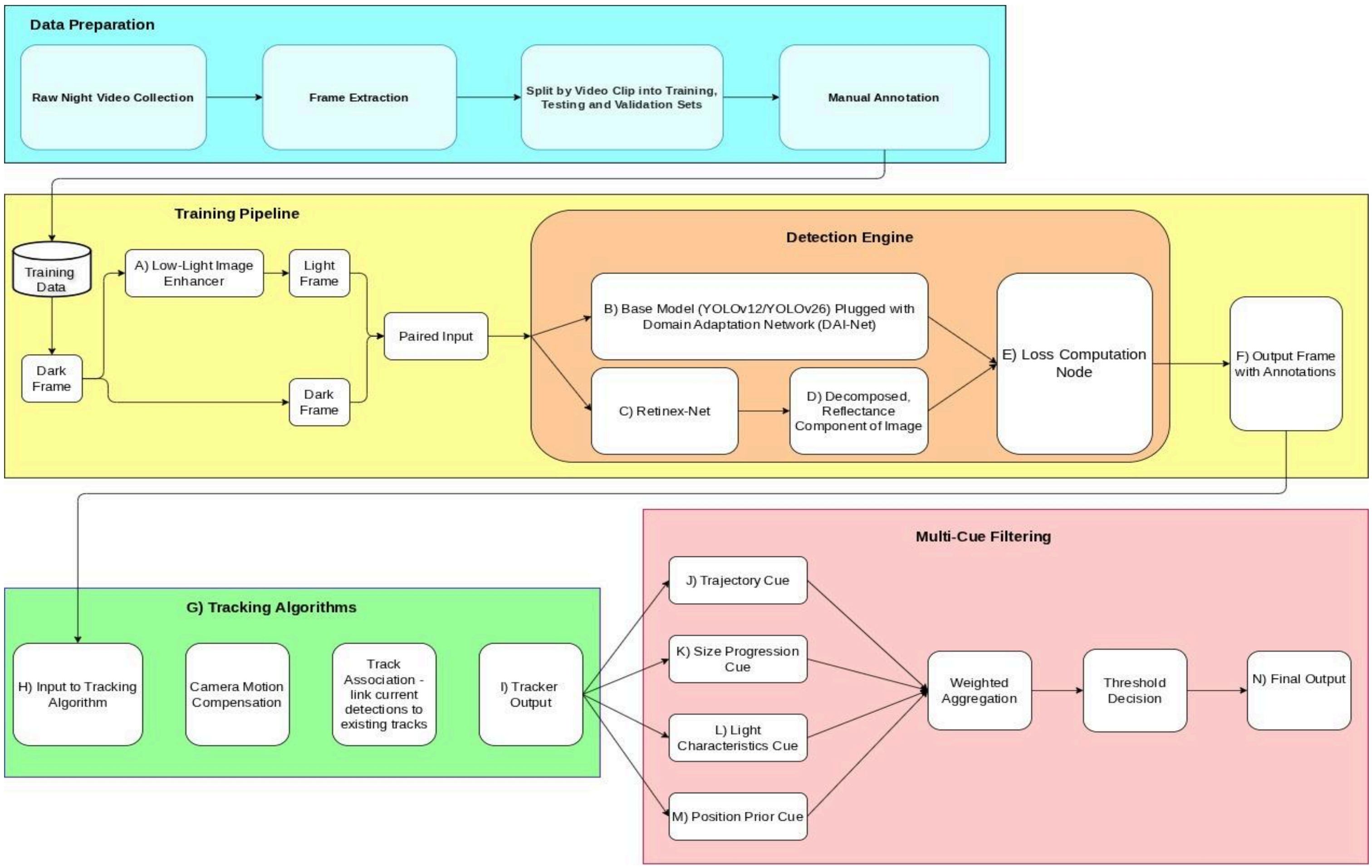
Nighttime CV Proposed System Pipeline



Links to the papers of the methods listed:

1. Zero-DCE: https://arxiv.org/pdf/2001.06826
2. Zero-DCE++: https://arxiv.org/pdf/2103.00860
3. DAI-Net: https://arxiv.org/pdf/2312.01220

Following is a short description of each high-level module in the proposed pipeline.

1. Enhancer Module

Purpose: Produce enhanced images to pair with the dark images from our dataset

Limitations: It is possible that the enhanced images turn out to be unsuitable for downstream vision tasks and result in poor accuracy downstream. Although chances are less since proposed technique has been shown to work well for downstream vision tasks.

Solution: Try out various other techniques. In this case, it is also necessary to ascertain whether the idea of using enhanced images itself might be causing a flaw. If so, rethinking of the entire pipeline might be required.

2. Detection Engine

Purpose: Take dark/light paired images as input, feed it to the base model plugged in with the domain adaptation model. It will output an annotated dark image that will be passed on to the tracking algorithm.

Limitations: The output image is expected to have a high number of false positives since no steps have been taken to reduce this number. The focus here is simply on getting as high as recall value as possible.

Solution: This situation will be taken care of by the combination of the tracking algorithm and the multi-cue filter, which are expected to filter out all false-positives, retaining only actual streetlight detections. It is highly unlikely that a situation will arise where the model still fails to make enough detections.

3. Tracking

Purpose: Track the objects detected by the model frame-by-frame. The main job is to prevent double-counting and saving the trajectory history of each unique object which is required by the next module.

Limitations: Extremely erratic camera motion might reduce the effectiveness of the tracking algorithm, making it more difficult to deduplicate results and accurately track the motion trajectory.

Solution: I propose to use tracking algorithms like UCMC track (2024) or Bot-SORT (inbuilt in Ultralytics), which take into account camera movements and make adjustments for that. Deeper study of both techniques is required which might enable to come up with ideas to tune it in case they don’t perform well out-of-the-box

4. Multi-Cue Filtering

Purpose: This is one of the most crucial parts. It will be used to filter out false-positives and increase precision. Various methods (listed in the descriptions below) will be used. For each filter that a predicted object passes, an increment will be made to its confidence score. At the end, all objects with a confidence score above a threshold will be accepted.

Limitations: Out of the proposed filtering techniques, the efficacy of each for our purpose varies. How effective a particular technique is will have to be found out empirically.

Solution: Less effective techniques will either have to be discarded or assigned lower weights. Also, there will arise a need to come up with more ways to filter out false-positives to replace weaker techniques.

Given below are more detailed explanations of smaller components of the proposed system.

A) Low-Light Image Enhancer

- This is one of the novel approaches for such a problem. The original domain adaptation network worked on the assumption that night-time data was unavailable. Thus, it used an algorithm to generate synthetic night-time images.
- Ours is a converse problem, which makes it unique. We will take a frame from a night-time video, convert it into a “day-time” image and then calculate the mutual feature alignment (MFA) loss between the two.
- Whether the MFA loss is helpful or not depends on the quality of the enhanced image. It has been proven that there is a disconnect between metrics that measure the enhancement of an image and whether that image actually performs well in a downstream task. This is because current evaluation metrics only optimize for the image to be visually appealing to human eyes.
- Since there is currently no metric that measures whether an image enhanced by a particular method will perform well on a downstream task, we must rely on data from previous experiments.
- A method called Zero-DCE has been shown to generate images which are not as visually appealing, but tend to perform well in downstream tasks, especially object detection. Although this paper was published in 2020, I am yet to come across a method which performs better for this particular task. I propose to use a newer variant - Zero-DCE++ (2021) - which gives similar or better results at a fraction of the computational cost. The minimal data and resource requirements of this method suit our project.

B) Base Model + Domain Adapter

- I propose to shortlist either YOLOv12 or YOLOv26 as the base model for our detector. Both are SOTA models with best-in-class accuracy. YOLOv26 has the advantage of being specifically developed to be lightweight and easier to deploy on low-resource devices. However, the model is new and unstable which might make it slightly challenging to work with, as compared to the slightly older and established YOLOv12. More reading about both these models is required from my side before making a decision.
- The model needs to be such that it is easy to prune/quantify it.
- For the domain adapter, I propose to use a version of DAI-Net. Its paper, published in CVPR 2025, boasts a significantly higher performance as compared to other SOTA techniques on the ExDark dataset. As mentioned above, using a night-to-day enhancer instead of day-to-night is one change we will be making to this technique.
- This method gets inspiration from Retinex theory and relies on decomposing images into their reflectance and illuminance components. It argues that since the reflectance is independent of illumination and never varies, training the detector simply on the reflectance part will bring better results.
- Apart from the MFA loss mentioned above, DAI-Net forces the earlier layers of the base model to produce feature maps similar to the one that would be produced for the reflectance component of the same image.
- This it does by comparing the feature map produced by the lower layers with the ground truth feature map of the reflectance layer. Where the ground truth feature map is obtained, is explained next.
- Since in our case, streetlight is a source of light itself, its visual signal in the decomposed reflectance image might be very small. Nevertheless, the reason I propose to still keep this component, is that this process might provide important contextual information for our base model, increasing its accuracy. If ablation studies find that no such advantage is obtained, we can remove this component altogether.

C) Retinex Net

- In order to decompose an image into its reflectance and illuminance components, and also produce a feature map for it, we need a learned network.
- Retinex Net is a light-weight pretrained network that does the job for us.
- There might be other such networks available. This is the one used by DAI-Net. I have not yet explored this aspect. If I can find some other similar network that works better for our purpose, we can use it. It will be another major change to the existing technique.

D) Decomposed, Reflectance Component of an Image

- This will be the output from the Retinex Net above.

- Notice, that this is not actually the ground truth. These are actually pseudo-labels which we will be presenting to the model as ground truth.
- The DAI-Net paper claims that making this assumption does not affect overall performance.

E) Loss Computation Node

- This is where the total loss is calculated.
- The paper proposes to calculate a redecomposition coherence loss. This is calculated by applying the “interchange-redecomposition-coherence” procedure to the decomposed low-light/well-lit reflectance generated by Retinex, by interchanging and reconstructing the images and then redecomposing them.
- The paper claims that this ensures the learned reflectance recomposition is consistent, stable and accurate.
- The MFA loss from earlier is added to this loss along with the regular detection loss to produce the combined loss.

F) Output Frame with Annotations

- This is the output of the previous blocks. It contains all the bounding boxes proposed by the model.
- We are looking to increase the recall as much as possible therefore this annotated output is likely to contain a lot of false positives, which will be accounted for in the subsequent modules.

G) Tracking Algorithms

- The camera will be mounted to a vehicle and will be moving. It is also likely that there might be a lot of shaking due to uneven road conditions and potholes.
- Algorithms need to be chosen that compensate for camera movement. Bot-SORT is a popular algorithm which comes included in Ultralytics and is easy to implement. UCMC Track is another novel approach which has proved to be effective for such tasks. UCMC track itself uses various motion cues for its tracking, so some of those cues could be exploited to be used in our next section

H) Input to Tracking Algorithm

- This will be the annotated bounding boxes from the current dark, annotated frame which is the output of the previous module.
- The input will also consist of the track history from the previous frames.

I) Tracker Output

- The output will contain bounding boxes as well.
- But here, each bounding box will be associated with an ID and the output will also contain the trajectory history of that particular object.

J) Trajectory Cue

- The streetlight is stationary with respect to its surroundings. When a vehicle moves down a road, the motion of the streetlights through the video is of a certain kind.
- The modeling of the trajectory of a light source as it moves across the frames is one cue to detect a streetlight

K) Size Progression Cue

- The area of the bounding box of the detected streetlight is expected to increase monotonically as the vehicle approaches the light. This could be used as another cue. This, however, could be a weaker cue which can be determined experimentally.

L) Light Characteristics Cue

- A streetlight has a certain appearance that is distinct from those coming from windows or shop-boards
- Windows and billboards have a diffused appearance whereas a streetlight has a near-saturated core with a radial halo

M) Position Prior Cue

- Streetlights are expected to appear in certain sections of the frames, specifically the upper sections.
- The place where the object first originated on the screen could give us some information about the object.
- We could also completely ignore a certain percentage of the bottom portion of the frame - since streetlights are very unlikely to be there - which will reduce a lot of the noise by removing headlights and some shop lights from the picture.

N) Final Output

- It is an image with confirmed streetlight detections with tracking IDs.

- This is where we can also classify the streetlight as being working/dim/not-working. This can be done simply on the basis of the luminance of the detected region and does not need a learned model.